

2025 HKDSE Physics

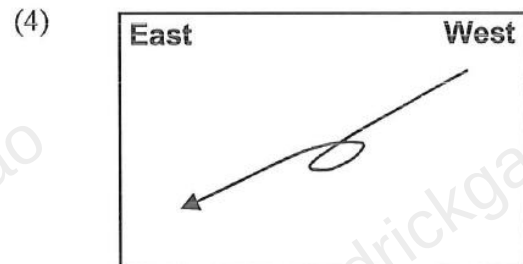
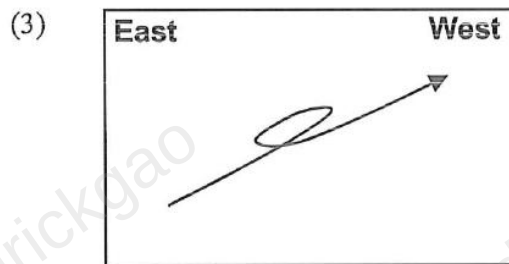
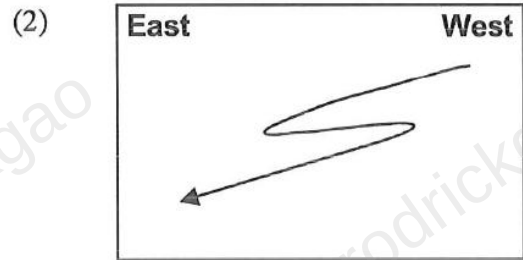
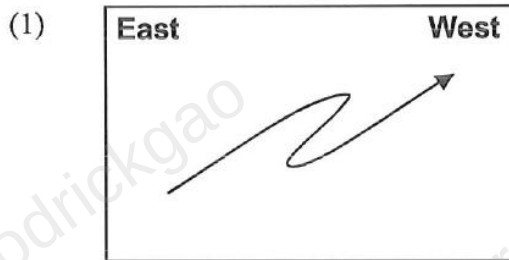
Paper 2 — Elective Sections

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Section A — Astronomy and Space Science

Question 1.1

Which of the following paths of a planet viewed against distant background stars show(s) retrograde motion?



Question 1.1 paths

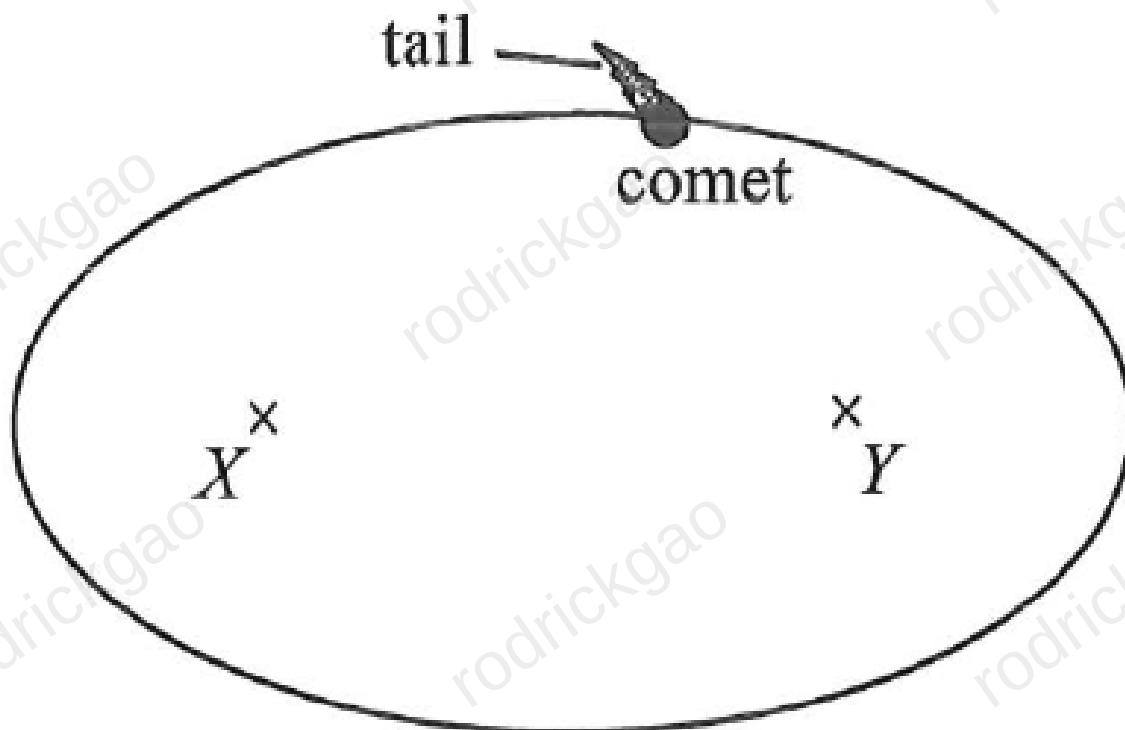
- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (4) only

Official answer: D

Retrograde motion must contain a loop or reversal relative to the normal background motion; diagrams (2) and (4) show this. Answer: D.

Question 1.2

A comet revolving about the Sun takes an elliptical orbit with foci X and Y as shown. It is known that the tail of a comet is always directed away from the Sun.



Question 1.2 comet

Which of the following descriptions about the comet **MUST** be correct?

- (1) The comet is travelling in a clockwise direction.
- (2) The comet attains its greatest speed when it is nearest to Y.
- (3) The gravitational potential energy of the comet in the orbit remains unchanged.

A. (1) only

B. (2) only

C. (1) and (3) only

D. (2) and (3) only

Official answer: B

A comet's tail points away from the Sun, so the Sun is at Y. Its speed is greatest at perihelion, so only (2) is correct; direction of motion and potential-energy change cannot be inferred as stated. Answer: B.

Question 1.3

Star Y is measured to have a parallax angle of p from telescopes on the Earth's surface. Suppose star Y is measured from telescopes on the surface of Mars (orbital radius 1.5 AU), the parallax angle would be

- A. larger than p .
- B. equal to p .
- C. smaller than p .
- D. either smaller, larger or equal to p .

Official answer: A

Parallax angle is proportional to the observing baseline. Mars has orbital radius 1.5 AU, so its baseline is larger and the parallax angle exceeds p . Answer: A.

Question 1.4

Star	Luminosity	Intensity observed	Colour
X	5	3	white
Y	2	4	orange
Z	3	1	blue

The table shows some properties of stars X, Y and Z. The luminosity and intensity observed are given respectively in the same arbitrary units. Which star is (1) the hottest; and (2) seen to be the brightest?

- A. (1) hottest: Y; (2) brightest: Z
- B. (1) hottest: Z; (2) brightest: Y
- C. (1) hottest: Z; (2) brightest: X
- D. (1) hottest: X; (2) brightest: Y

Official answer: B

The blue star Z has the highest surface temperature, while Y has the greatest observed intensity. Answer: B.

Question 1.5

Stars X and Y have the same surface temperature. The luminosities of X and Y are $L_X = 121000 L_s$ and $L_Y = 0.1 L_s$ respectively, where L_s is the luminosity of the Sun. Find $R_X : R_Y$, where R_X and R_Y are radii of X and Y respectively.

- A. 1210000 : 1
- B. 1 : 1210000
- C. 1100 : 1
- D. 1 : 1100

Official answer: C

$L = 4\pi R^2 \sigma T^4$. At equal temperature, $R \propto \sqrt[4]{L}$, so $R_X : R_Y = \sqrt[4]{(121000/0.1)} = 1100:1$. Answer: C.

Question 1.6

The spectral class of a main-sequence star can be determined by

- (1) the surface temperature of the star.
- (2) the colour of the star.
- (3) the intensity of the lines in the stellar spectrum.

A. (1), (2) and (3)

B. (1) and (3) only

C. (2) only

D. (1) only

Official answer: A

Spectral class is determined by surface temperature and is reflected in stellar colour and absorption-line strengths. Statements (1), (2) and (3) are correct. Answer: A.

Question 1.7

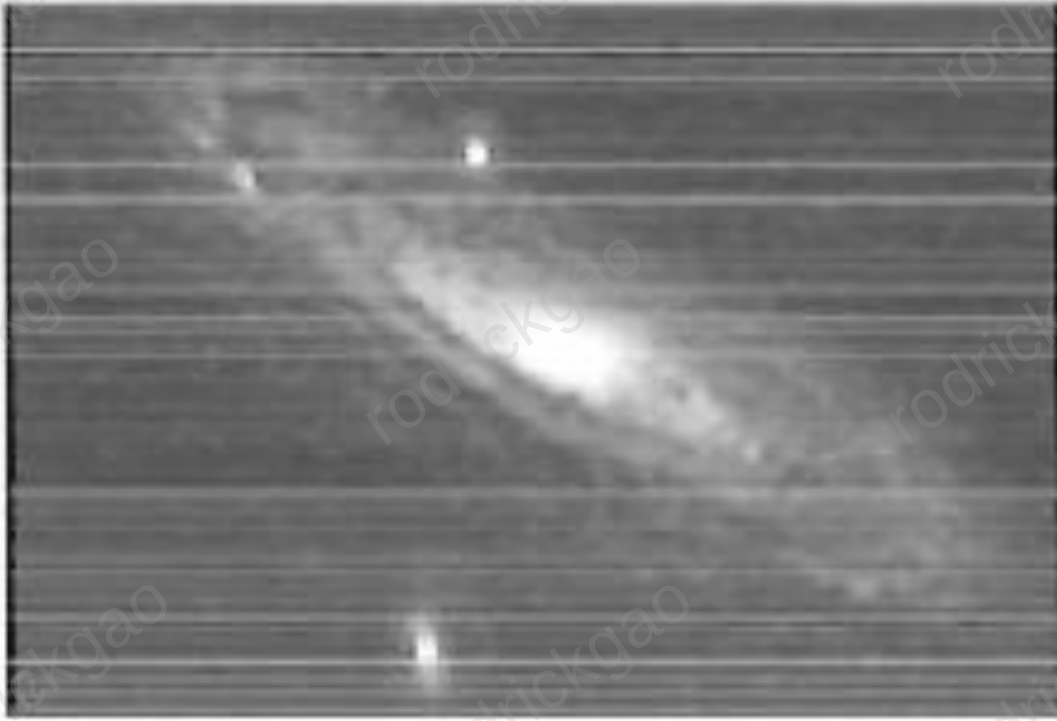
In the Hertzsprung-Russell (H-R) diagram, stars of the same apparent magnitude occupy

- A. the same horizontal line.
- B. the same vertical line.
- C. the same diagonal going from lower-right to upper-left.

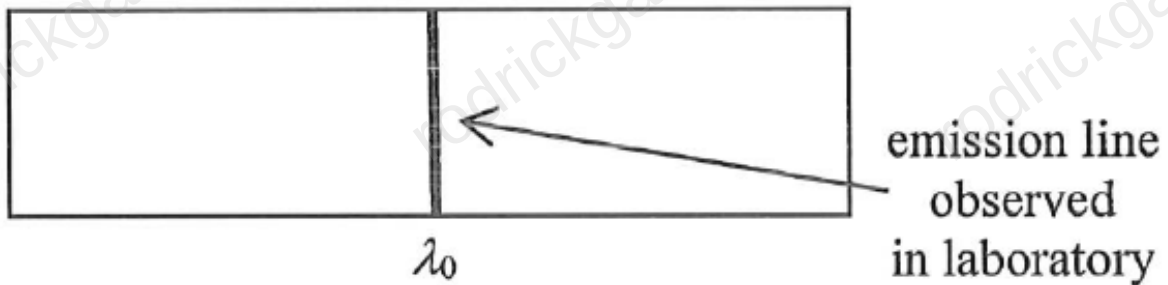
D. arbitrary locations in the diagram.

Official answer: D

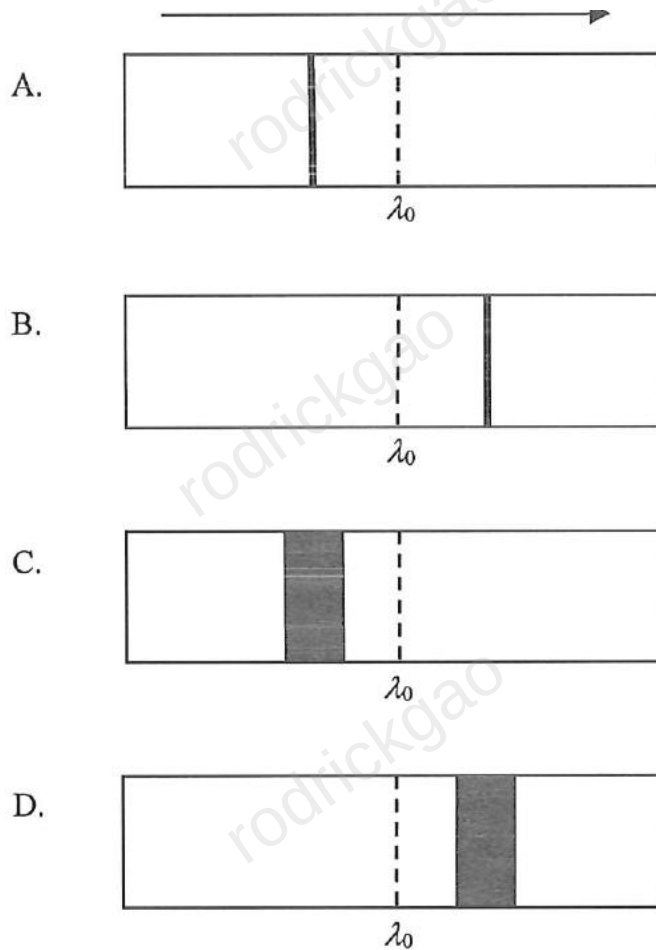
An H-R diagram plots absolute luminosity against surface temperature. Apparent magnitude also depends on distance, so stars with the same apparent magnitude may occupy any region. Answer: D.

Question 1.8*Question 1.8 galaxy*

Which of the following best shows the observed emission line (rest wavelength λ_0 observed in laboratory) of the above blue-shifted spiral galaxy?



Question 1.8 laboratory spectrum
increasing wavelength



Question 1.8 graphical options

Official answer: C

The galaxy's overall blue shift places the line centre to the left of λ_0 ; rotation gives different radial velocities across the galaxy, broadening the line into a band. Answer: C.

Structured Question 1

Given: mass of the Moon $M_m = 7.34 \times 10^{22}$ kg; radius of the Moon $R_m = 1.74 \times 10^6$ m

To set up a permanent lunar station on the Moon, one has to overcome the issue of transporting astronauts/supplies between the Earth and the lunar station.

The following two-stage return scheme is proposed for bringing astronauts from the Moon back to the Earth:

- (I) An Ascent Vehicle S_A is launched from the Moon's surface to dock with an Orbiting Vehicle S_B which is orbiting circularly around the Moon at an altitude H above the Moon's surface.
- (II) The Orbiting Vehicle carrying the astronauts then returns to the Earth.

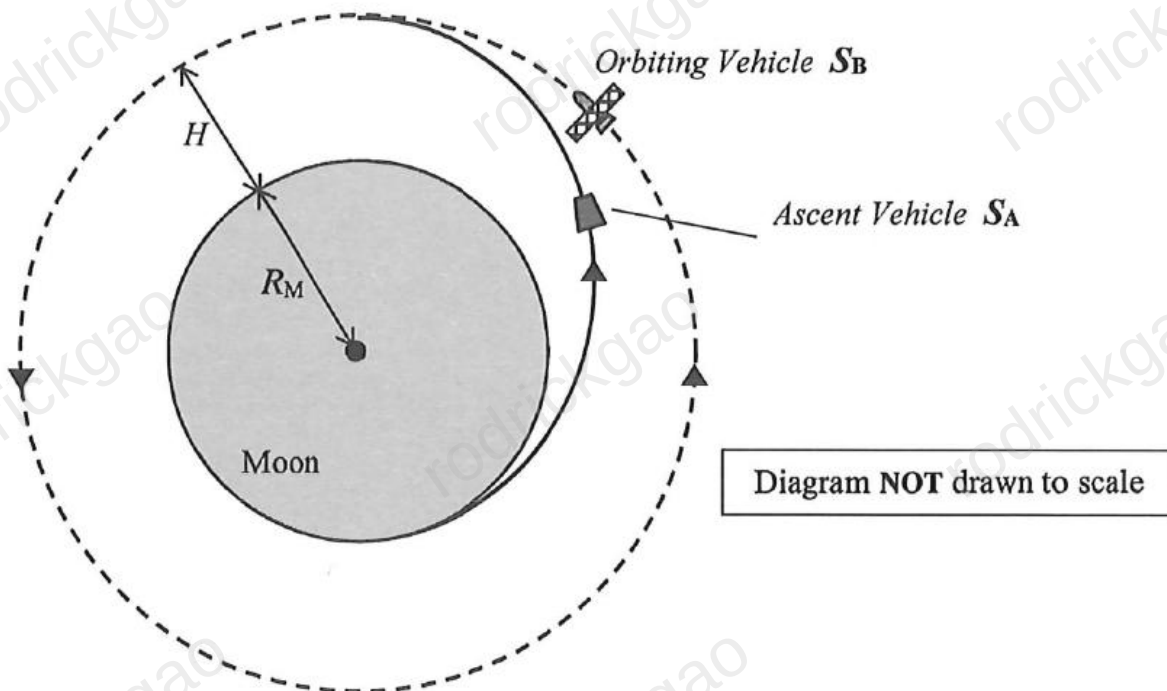


Figure 1.1

You may neglect the gravitational effects of the Earth in your calculation. Given: $H = 0.20 \times 10^6 \text{ m}$

- (a) Find the orbital speed v_0 of S_B , which is close to 1.6 km s^{-1} . Give your answer correct to 3 significant figures.

(3 marks)

- (b) An astronaut inside S_B experiences weightlessness. A student suggests that this is because the Moon's gravitational force acting on the astronaut is zero. Explain whether this is correct or not. (2 marks)

In planning the return journey, the Ascent Vehicle S_A launched at velocity v_1 from the Moon's surface will then dock with the orbiting S_B , where the two space vehicles are moving with the same speed v_0 (i.e. the orbital speed of S_B) when docking.

- (c) (i) Determine the launching velocity v_1 required. Assume that the rocket engine of S_A is shut immediately after it has attained v_1 on the Moon's surface. (3 marks)
- (ii) Referring to the escape velocity $v_e = 2.38 \text{ km s}^{-1}$ of the Moon, explain, regarding the design of S_A , an advantage of this two-stage return scheme compared to a direct return flight. (2 marks)

Official marking scheme

Solution		
1. (a)		
$\frac{mv_o^2}{(R_M + H)} = \frac{GM_M m}{(R_M + H)^2}$	1M	
$v_o = \sqrt{\frac{GM_M}{R_M + H}}$	1M	
$= \sqrt{\frac{(6.67 \times 10^{-11})(7.34 \times 10^{22})}{1.74 \times 10^6 + 0.20 \times 10^6}}$	1A	Accept: $1.57 \sim 1.60 \text{ km s}^{-1}$
$= 1.588584 \times 10^3 \text{ m s}^{-1} \approx 1.59 \text{ km s}^{-1}$	3	
(b)		
This is incorrect. In orbiting the Moon, the astronaut of mass m experiences a non-zero gravitational force $F = \frac{GM_M m}{(R_M + H)^2}$	1A	Or All the gravitational force is used as the centripetal force.
while both S_B and the astronaut inside are in free fall so that the reaction force between them is zero.	1A	
	2	
(c) (i)		
By conservation of energy:		
$\frac{1}{2}mv_1^2 - \frac{GM_M m}{R_M} = \frac{1}{2}mv_o^2 - \frac{GM_M m}{R_M + H}$	1M	
$v_1^2 = v_o^2 + 2GM_M \left(\frac{1}{R_M} - \frac{1}{R_M + H} \right)$		
$= 1589^2 + 2(6.67 \times 10^{-11})(7.34 \times 10^{22})$	1M	e.c.f. from (a)
$\left(\frac{1}{1.74 \times 10^6} - \frac{1}{1.74 \times 10^6 + 0.20 \times 10^6} \right)$		
$= 3.103737 \times 10^6$		
$v_1 = 1.761742 \times 10^3 \text{ m s}^{-1} = 1.76 \text{ km s}^{-1}$	1A	Accept: $1.76 \text{ km s}^{-1} \sim 1.80 \text{ km s}^{-1}$
	3	
(ii)		
It requires a smaller launching velocity ($v_1 = 1.76 \text{ km s}^{-1} < v_e = 2.38 \text{ km s}^{-1}$) for S_A to be sent from the lunar surface,	1A	e.c.f. from (c)(i)
thus a less powerful / less massive engine / less fuel is needed.	1A	
	2	

(a) Let $r = R_M + H$. From $GM_M m / r^2 = mv_o^2 / r$, $v_o = \sqrt{[GM_M / (R_M + H)]} = 1.59 \text{ km s}^{-1}$.

(b) The statement is false. Lunar gravity is non-zero and supplies the centripetal force; the astronaut and spacecraft fall freely together, so the support force between them is zero and the astronaut feels weightless.

(c) (i) $\frac{1}{2}mv_1^2 - GM_M m / R_M = \frac{1}{2}mv_o^2 - GM_M m / (R_M + H)$, giving $v_1 = 1.76 \text{ km s}^{-1}$.

(ii) Since $v_1 < \text{escape speed } 2.38 \text{ km s}^{-1}$, less energy, fuel and engine power / mass are required.

Section B — Atomic World

Question 2.1

In Thomson's plum pudding model, it is suggested that an atom is a positively charged sphere with electrons distributed evenly within it. Which of the following results of the α -scattering experiment implies that Thomson's model is incorrect?

- (1) Most of the α particles pass straight through the thin gold foil.
- (2) Some α particles are deflected by large angles.
- (3) A few α particles are bounced back.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: D

Large-angle deflection and backscattering require positive charge and mass to be concentrated in a tiny nucleus, contradicting a uniform positive sphere. Statements (2) and (3) are correct. Answer: D.

Question 2.2

Which of the following about the Bohr atomic model is/are correct?

- (1) Electrons are orbiting around the nucleus.
- (2) The attractive force between the nucleus and the electron provides the centripetal force for the electron's circular motion.
- (3) Photons are emitted when electrons jump from lower energy orbits to higher energy orbits.

- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only

Official answer: C

In the Bohr model, an electron orbits the nucleus with electrostatic force providing centripetal force. Light is emitted only in a transition from a higher to a lower level. Hence (1) and (2). Answer: C.

Question 2.3

The energy levels of a hydrogen-like atom take the form $-36/n^2$ eV, where n is the quantum number. What is the energy level of its 2nd excited state?

- A. -4 eV
- B. -9 eV
- C. -12 eV
- D. -18 eV

Official answer: A

The second excited state is $n = 3$. Thus $E_3 = -36/3^2 = -4$ eV. Answer: A.

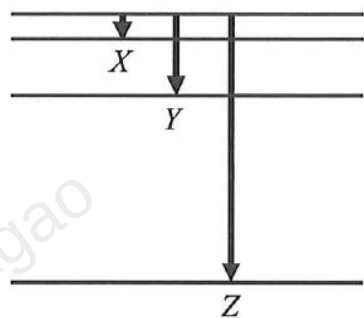
Question 2.4

increasing frequency

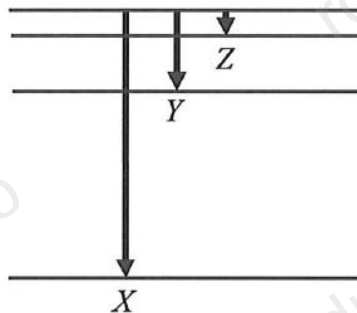
*Question 2.4 spectrum*

The above figure shows the spectrum obtained when the visible light from a discharge tube passes through a diffraction grating. Spectral lines X, Y and Z correspond to electron transitions between four energy levels. Which of the following best illustrates the electron transitions that give the spectral lines?

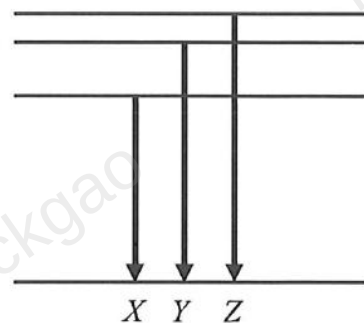
A.



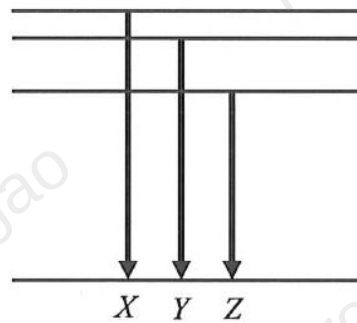
B.



C.



D.



A



B



C



D

*Question 2.4 graphical options***Official answer: C**

Since $f \propto \Delta E$, Z has the greatest frequency, Y is slightly lower and X is the lowest. The three shown transitions to the ground state have successively larger energy gaps. Answer: C.

Question 2.5

A beam of fast-moving electrons of velocity v behaves like a wave with wavelength λ and frequency f . Which of the following statements is/are correct?

- (1) λ increases when v decreases.
- (2) The beam of electrons shows wave properties only with some discrete f .
- (3) The momentum associated with the electron beam is given by h/λ , where h is the Planck constant.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: C

$\lambda = h/(mv)$, so reducing v increases λ . Matter-wave frequency is not restricted to discrete values, and $p = h/\lambda$. Statements (1) and (3) are correct. Answer: C.

Question 2.6

Two point objects of separation 2.4×10^{-6} m are to be observed under a microscope using light of wavelength 550 nm. Which of the following combinations of aperture diameter and focal length of the microscope's objective lens is suitable?

aperture diameter focal length

- A. 5 mm 20 mm
- B. 6 mm 20 mm
- C. 5 mm 24 mm
- D. 6 mm 24 mm

Official answer: B

Resolution requires object angular separation $s/f \geq 1.22\lambda/D$. Comparing the four cases, only $D = 6$ mm and $f = 20$ mm satisfies the condition. Answer: B.

Question 2.7

Which of the following statements about a scanning tunnelling microscope (STM) is/are correct?

- (1) It uses a high-energy electron beam instead of visible light.
- (2) STM gives a 3-dimensional image of a conductive surface.
- (3) The tunnelling current produced depends on the separation between the surface under investigation and the probe of the STM.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: D

An STM uses tunnelling current, not a high-energy electron beam. It gives a three-dimensional image of a conducting surface, and the current depends strongly on tip distance. Statements (2) and (3) are correct. Answer: D.

Question 2.8

When a material is broken up into nanosized particles, which of the following statements is/are correct?

- (1) Its chemical properties change but the physical properties remain unchanged.
- (2) Its shape changes but the atomic arrangement remains unchanged.
- (3) Its total surface area to volume ratio increases.

- A. (1) only
- B. (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: B

At nanoscale, material properties and atomic arrangement need not remain unchanged. The definite increase is total surface-area-to-volume ratio; only (3) is correct. Answer: B.

Structured Question 2

- (a) Figure 2.1 shows the lowest four energy levels of a hydrogen atom.

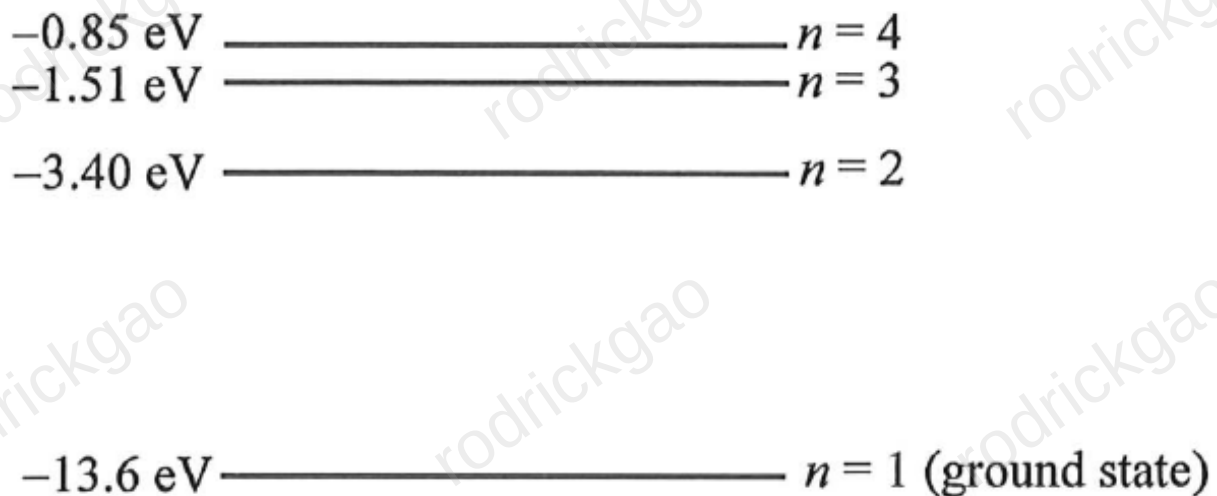


Figure 2.1

When a hydrogen atom in ground state and a photon of energy 10.2 eV collide,

- state all subsequent changes to the hydrogen atom. (2 marks)
 - find the wavelength of the photon subsequently emitted by the hydrogen atom. (2 marks)
- (b) The electrode X of the photocell shown below is made of zinc and it is illuminated by photons of energy 10.2 eV . Given: work function of zinc is 4.9 eV

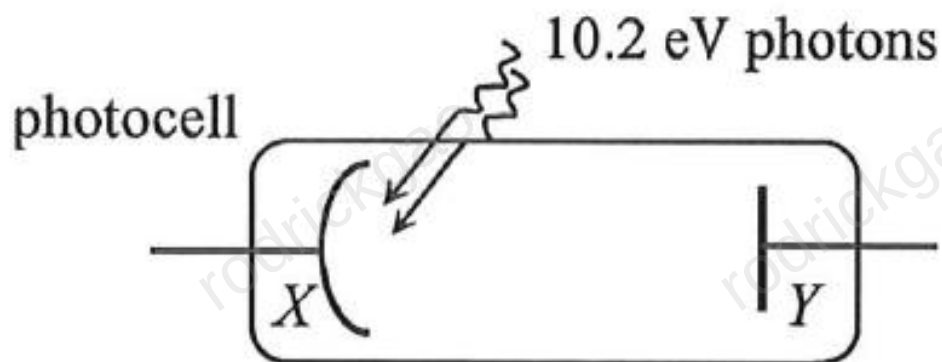
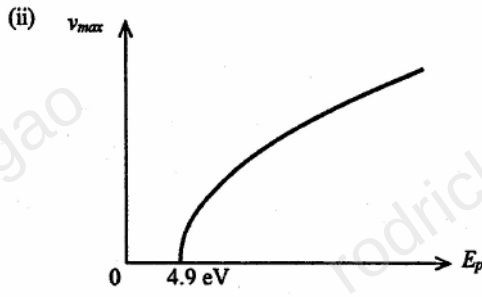


Figure 2.2

- Find the maximum speed v_{max} of the photoelectrons emitted from electrode X. (2 marks)
- In your answer book, sketch a graph (including the origin) showing how this maximum speed v_{max} depends on the energy E_p of the incident photons. (2 marks)
- Explain why 10.2 eV photons are able to remove electrons from the zinc surface whereas they can only excite hydrogen atoms in ground state without releasing the electron from the atom. (2 marks)

Official marking scheme

Solution	Marks	Remarks
2. (a) (i) The atom absorbs the photon and excites to the first excited state / energy level with $n = 2$ / $E = -3.40$ eV and subsequently falls back to the ground state / releasing a (10.2 eV) photon.	1A 1A 2	
(ii) $\lambda = \frac{hc}{\Delta E}$ $= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{10.2 \times (1.60 \times 10^{-19})}$ $= 1.21875 \times 10^{-7} \text{ m}$ $\approx 1.22 \times 10^{-7} \text{ m}$	1M 1A 2	Accept: $1.20 \times 10^{-7} \sim 1.22 \times 10^{-7} \text{ m}$
(b) (i) Maximum kinetic energy $\frac{1}{2} m_e v_{\max}^2 = (10.2 - 4.9) \text{ eV}$ $\frac{1}{2} (9.11 \times 10^{-31}) v_{\max}^2 = 5.3 \text{ eV}$ $v_{\max} = \sqrt{\frac{2 \times 5.3 \times 1.60 \times 10^{-19}}{9.11 \times 10^{-31}}}$ $= 1.364438 \times 10^6 \text{ m s}^{-1} \approx 1.36 \times 10^6 \text{ m s}^{-1}$	1M 1A 2	Accept: $1.36 \times 10^6 \sim 1.40 \times 10^6 \text{ m s}^{-1}$
(ii) 	2A 2	
(c) As the photon energy (10.2 eV) is larger than the work function of zinc (4.9 eV), it is enough to liberate electrons from the zinc surface. However, the photon energy (10.2 eV) is smaller than the ionization energy of hydrogen (13.6 eV). It cannot release the electron from the hydrogen atom in ground state.	1A 1A 2	

(a) (i) 10.2 eV equals the $-13.6 \rightarrow -3.4$ eV gap, so the atom absorbs the photon to reach $n = 2$, then returns to the ground state and emits a 10.2 eV photon.

(ii) $\lambda = hc/\Delta E = 1.22 \times 10^{-7} \text{ m}$.

(b) (i) $K_{\max} = (10.2 - 4.9) \text{ eV} = 5.3 \text{ eV}$; $v_{\max} = \sqrt{2K/m_e} = 1.36 \times 10^6 \text{ m s}^{-1}$.

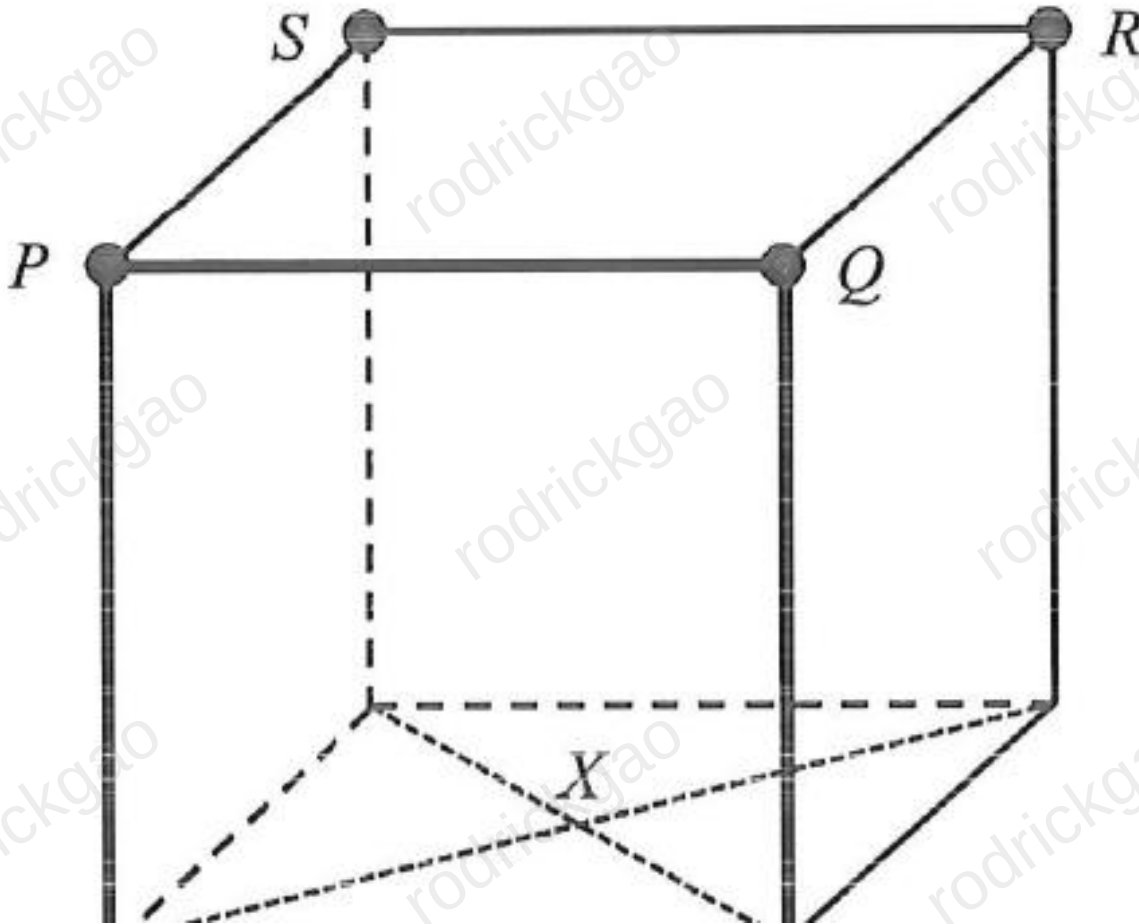
(ii) $v_{\max} = \sqrt{2(E_p - 4.9 \text{ eV})/m_e}$. There are no photoelectrons for $E_p < 4.9 \text{ eV}$; the curve starts at (4.9 eV, 0) and is concave downward.

(c) 10.2 eV exceeds zinc's 4.9 eV work function, so electrons are emitted, but it is below hydrogen's 13.6 eV ionisation energy and can only excite ground-state hydrogen.

Section C — Energy and Use of Energy

Question 3.1

In a room of dimensions $d \times d \times d$, identical point light sources P, Q, R and S are located at the four corners of the ceiling as shown. The luminous flux of each source is Φ .



Question 3.1 room

Find the illuminance around point X on the floor in the middle of the room. Neglect reflections from the walls of the room.

- A. $[\Phi/(\pi d^2)] \times (3/2)^{1/2}$
- B. $[\Phi/(\pi d^2)] \times (3/2)^{3/2}$
- C. $[\Phi/(\pi d^2)] \times (2/3)^{1/2}$
- D. $[\Phi/(\pi d^2)] \times (2/3)^{3/2}$

Official answer: D

For each lamp, $r = d\sqrt{3/2}$ and $\cos \theta = d/r = \sqrt{2/3}$. Therefore $E = 4[\Phi/(4\pi r^2)]\cos \theta = (\Phi/\pi d^2)(2/3)^{3/2}$. Answer: D.

Question 3.2

The end-use energy efficiency of a microwave oven is high because

- (1) microwaves only heat up the food without heating up the utensil directly.
- (2) microwaves do not heat up the nearby air much.
- (3) no energy is lost as heat in the circuit of the microwaves-generating component.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

Official answer: A

Microwaves mainly heat the food directly and heat the surrounding air only slightly; the generator circuit still dissipates heat. Only (1) and (2) are correct. Answer: A.

Question 3.3

Which of the following arrangements can reduce the Overall Thermal Transfer Value (OTTV) of a building envelope?

- (1) using air-conditioners which are more energy-saving
- (2) using glass windows with low-emissivity coating
- (3) using building materials with greater U-value

A. (1) only

B. (2) only

C. (1) and (3) only

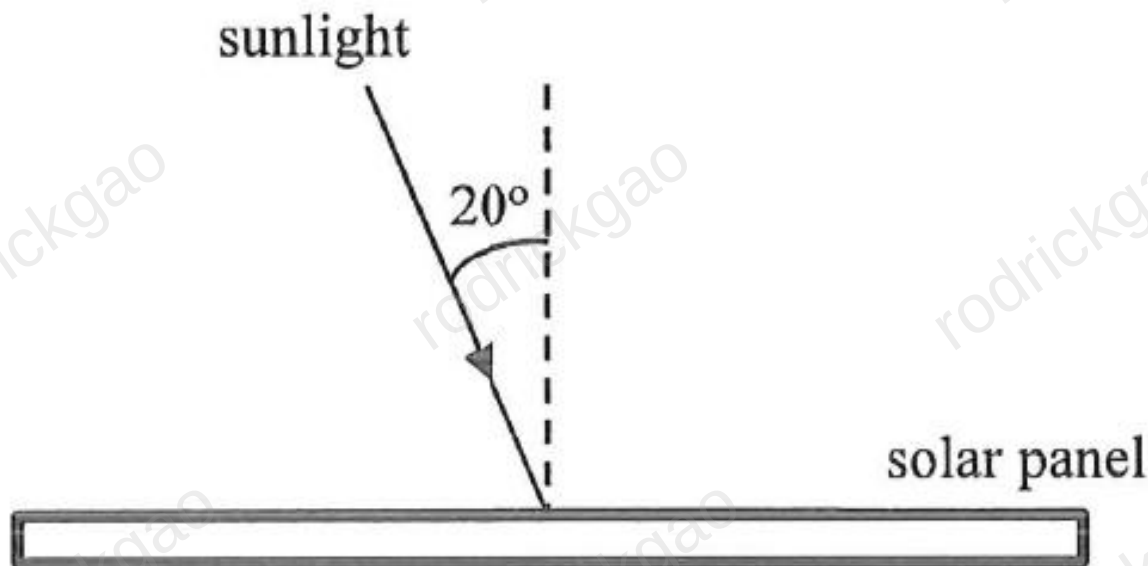
D. (2) and (3) only

Official answer: B

Low-emissivity glass reduces heat transfer. Air-conditioner efficiency does not change the envelope OTTV, while a larger U-value increases heat transfer. Answer: B.

Question 3.4

When incident sunlight makes an angle of 20° to the normal of a solar panel, the electric power generated is 100 W.



Question 3.4 solar panel

Assuming that the efficiency of the solar panel remains unchanged, find the electric power generated if the sunlight makes an angle of 30° to the normal of the panel and its intensity is reduced by 10%.

A. $100 \times (\sin 30^\circ / \sin 20^\circ) \times 0.1 \text{ W}$

B. $100 \times (\cos 30^\circ / \cos 20^\circ) \times 0.1 \text{ W}$

C. $100 \times (\sin 30^\circ / \sin 20^\circ) \times 0.9 \text{ W}$

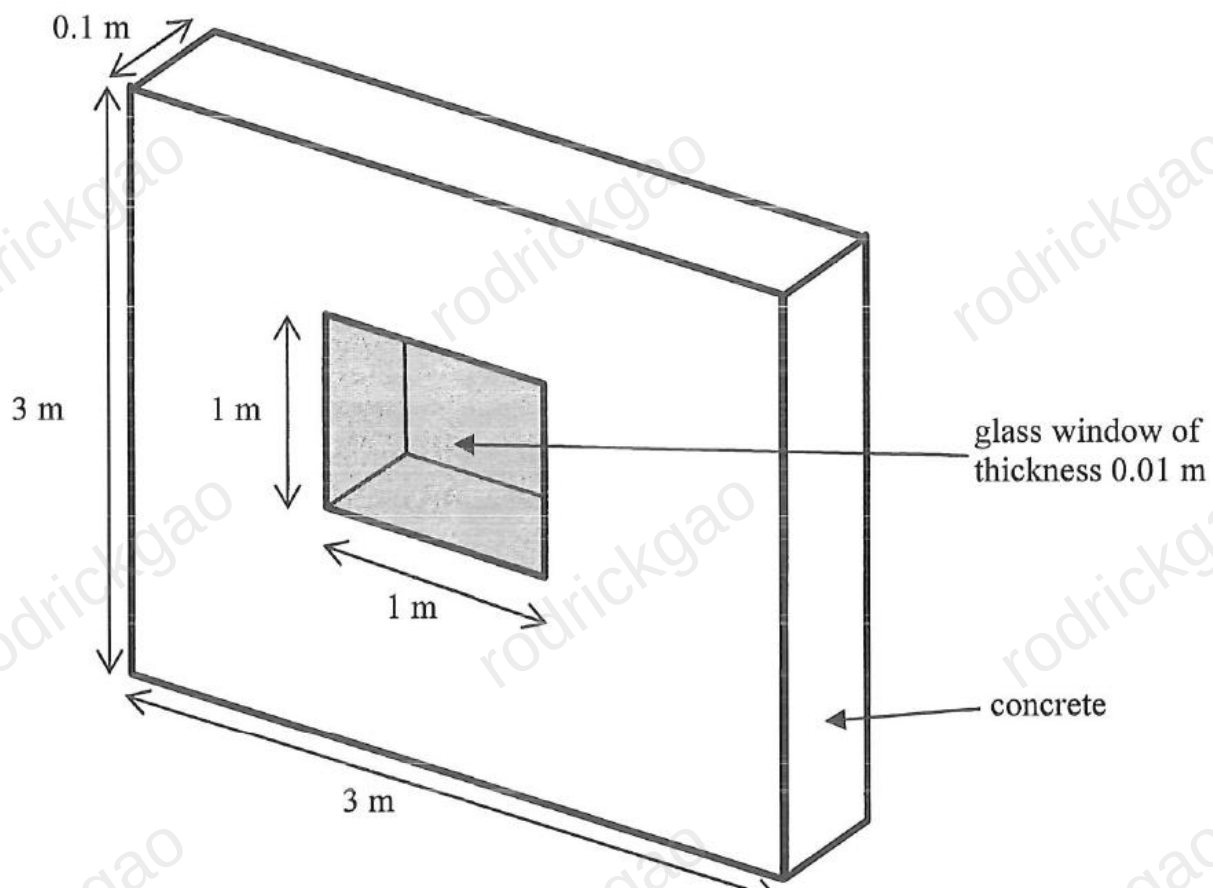
D. $100 \times (\cos 30^\circ / \cos 20^\circ) \times 0.9 \text{ W}$

Official answer: D

Incident power is proportional to $I \cos \theta$. Hence $P_2 = 100 \times 0.9 \times \cos 30^\circ / \cos 20^\circ$. Answer: D.

Question 3.5

The figure shows a $3\text{ m} \times 3\text{ m}$ wall made of concrete of thickness 0.1 m and having a $1\text{ m} \times 1\text{ m}$ glass window of thickness 0.01 m .



Question 3.5 wall

The thermal conductivities of concrete and glass are $0.8\text{ W m}^{-1}\text{ K}^{-1}$ and $1.3\text{ W m}^{-1}\text{ K}^{-1}$ respectively. Find the overall thermal transmittance (U-value) of the wall.

A. $20\text{ W m}^{-2}\text{ K}^{-1}$

B. $22\text{ W m}^{-2}\text{ K}^{-1}$

C. $73\text{ W m}^{-2}\text{ K}^{-1}$

D. $116\text{ W m}^{-2}\text{ K}^{-1}$

Official answer: B

$U_c = 0.8/0.1 = 8$ and $U_g = 1.3/0.01 = 130\text{ W m}^{-2}\text{ K}^{-1}$. Thus $U = (8 \times 8 + 130 \times 1)/9 = 21.6 \approx 22\text{ W m}^{-2}\text{ K}^{-1}$. Answer:

B.

Question 3.6

Air-conditioners P and Q below operate respectively to cool two identical rooms under the same environmental conditions.

Air-conditioner	Input power / kW	Coefficient of performance (COP)
P	1.6	2.2
Q	2.0	1.9

If P takes 400 s to cool down the room, estimate the time required for Q to cool down the other room to the same extent.

- A. 688 s
- B. 432 s
- C. 371 s
- D. 276 s

Official answer: C

Cooling power = COP × input power: P supplies 3.52 kW and Q supplies 3.80 kW. Therefore $t_Q = 400 \times 3.52 / 3.80 = 371$ s. Answer: C.

Question 3.7

Wind turbines P and Q below deliver roughly the same electric power output when they operate under the conditions tabulated.

Wind turbine	Blade length / m	Wind speed / m s ⁻¹ (blowing normally to the turbine)
P	23	6.2
Q	15	8.0

The overall efficiency of electric power generation

- A. is roughly the same for both turbines.
- B. is larger for P.
- C. is larger for Q.
- D. cannot be compared because the maximum power available for tapping from the wind is not the same.

Official answer: C

Available wind power $\propto r^2 v^3$. P's $23^2 \times 6.2^3$ exceeds Q's $15^2 \times 8.0^3$, yet their outputs are equal; therefore Q is more efficient. Answer: C.

Question 3.8

What are the functions of pressurised water in the core of a pressurised water reactor?

- (1) It acts as a moderator.
- (2) It acts as a coolant.
- (3) It turns into steam and drives the turbine.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

Official answer: A

In a pressurised-water reactor, the pressurised primary water acts as moderator and coolant but does not boil. The turbine is driven by secondary-loop steam. Statements (1) and (2) are correct. Answer: A.

Structured Question 3

Indoor vertical farming is suitable for urban areas where land is limited. Figure 3.1 shows a container modified into a vertical farming system for growing plants under optimal conditions provided by artificial lighting and temperature control.

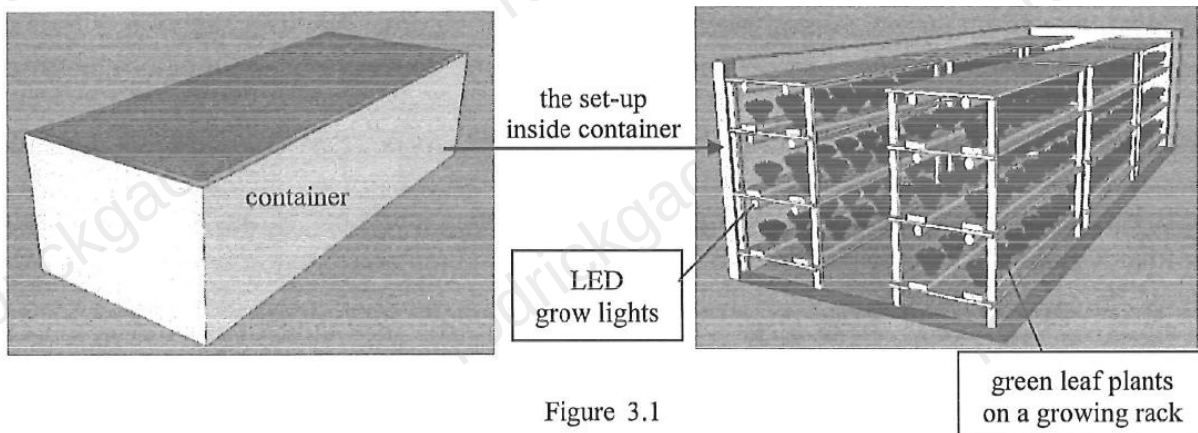


Figure 3.1

Figure 3.1

- (a) (i) State TWO reasons why light-emitting diodes (LEDs) are preferred to be used as grow lights rather than fluorescent tube lamps (FTLs). (2 marks)
- (ii) Taking into account energy efficiency, state which lighting system is the best for growing green leaf plants: (I) green LEDs, (II) red and blue LEDs and (III) full spectrum (white) LEDs. (1 mark)
- (b) Assume that all surfaces of the container are made from the same metal plate and that heat transfer through the floor is negligible. The container's interior temperature is maintained at an optimum value of 20 °C. The table below shows the variation of the interior and exterior temperatures of the container within 24 hours.

Time	LEDs lighting system	Interior temperature (°C)	Exterior temperature (°C)
8:00 a.m. – 8:00 p.m.	ON	20	32
8:00 p.m. – 8:00 a.m.	OFF	20	24

Given: total exposed surface area of the container = 70 m^2

U-value of the metal plate of the container = $12.5 \text{ W m}^{-2} \text{ K}^{-1}$

- Estimate the total thermal energy gain from outside the container during this 24-hour period. (2 marks)
- Given that the air-conditioning system consumes 0.3 J of electrical energy for removing 1 J of thermal energy from the container. Estimate the cost of electricity per day if the air-conditioning system is required to remove $6.45 \times 10^8 \text{ J}$ of thermal energy during this 24-hour period. Given: cost of electricity = \$1.4 per kWh (2 marks)
- Estimate the minimum cooling capacity of the container's air-conditioning system, in kW, required for maintaining the interior temperature at 20°C . Given that the total power of the lighting system is 1500 W and 60% of this input power eventually becomes thermal energy. (2 marks)
- Without changing the dimensions and material of the container, suggest one modification to the container so as to reduce the energy consumption. (1 mark)

Official marking scheme

Solution		Marks
3. (a) (i) Any TWO: - LEDs emit light / The light from LEDs can be easily directed in a specific direction (towards the plants) while FTLs emit light in all directions. - LEDs are more efficient. - LEDs have longer life-time / replacement of LED is less frequent. - LEDs have no toxic materials while FTLs contain mercury. - LEDs emitting a definite wavelength / colour are available. (ii) (II) Red and blue LEDs (b) (i) Q from outside $= 12.5 \times 70 \times [(24 - 20) + (32 - 20)] \times (12 \times 3600)$ $= 6.048 \times 10^8 \text{ J} \approx 605 \text{ MJ}$ (ii) Total electrical energy consumed by air-conditioning system $= \frac{6.45 \times 10^8 \times 0.3}{3600 \times 1000} = 53.75 \text{ kWh}$ Cost of electricity $= 53.75 \times 1.4 = \$75.25 \approx \75.3 (iii) Minimum cooling capacity $= 1500 \times 0.60 + [12.5 \times 70 \times (32 - 20)]$ $= 11400 \text{ W} = 11.4 \text{ (kW)}$ (c) Any ONE: - Install insulating materials on the roof / walls of the container. - Paint the container white outside. - Install solar panels / a lawn on the container's roof.	2A	
	2	
	1A	
	1	
	1M	
	1A	Accept: $6.00 \times 10^8 \sim 6.05 \times 10^8 \text{ J}$
	2	
	1M	
	1A	Accept: $\$75.0 \sim \75.5
	2	
	1M	
	1A	
	2	
	1A	
	1	

(a) (i) Any two: directional light, higher efficiency, longer life, mercury-free, or selectable wavelengths.

(ii) Choose red and blue LEDs in (II): green leaves mainly absorb red and blue, avoiding input at poorly absorbed green wavelengths.

(b) (i) $Q = UA[(32-20)+(24-20)](12 \times 3600) = 6.048 \times 10^8 \approx 6.05 \times 10^8 \text{ J}$.

(ii) $E_{el} = 0.3(6.45 \times 10^8) = 1.935 \times 10^8 \text{ J} = 53.75 \text{ kWh}$; cost $= 53.75 \times \$1.4 = \$75.25 \approx \$75.3$.

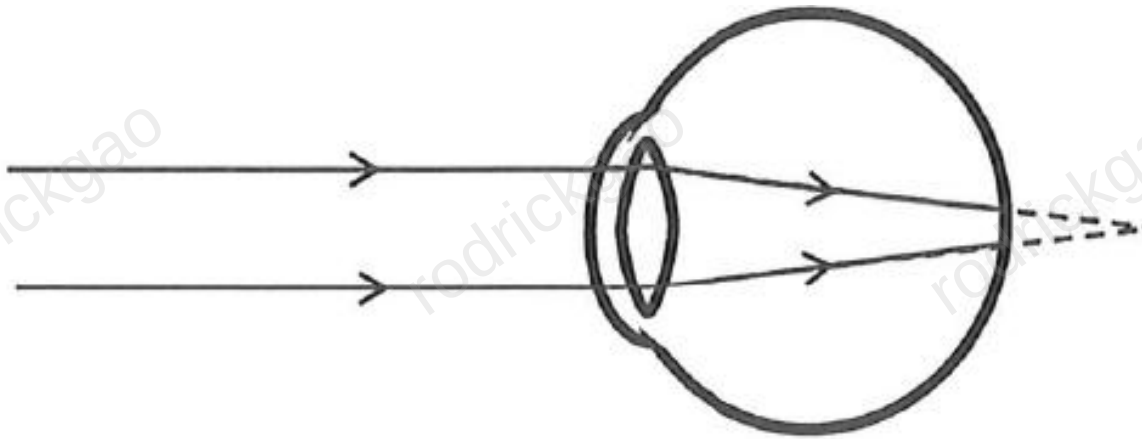
(iii) Peak cooling load $= 0.60(1500) + 12.5(70)(32-20) = 11400 \text{ W} = 11.4 \text{ kW}$.

(c) Add thermal insulation to the roof / walls; valid alternatives include a white exterior, solar panels or a green roof.

Section D — Medical Physics

Question 4.1

The figure shows an eye viewing a distant object.



Question 4.1 eye

Which of the following descriptions is correct?

Option — eye lens — can be corrected by

- A. too thin — a converging lens
- B. too thick — a converging lens
- C. too thin — a diverging lens
- D. too thick — a diverging lens

Official answer: A

Parallel rays focus behind the retina, showing insufficient converging power / a lens that is too thin. A converging lens corrects this. Answer: A.

Question 4.2

The human eye is most sensitive to light of wavelength 550 nm. Given that the diameter of the pupil of a normal eye is 5 mm, estimate the angular resolving power of the eye.

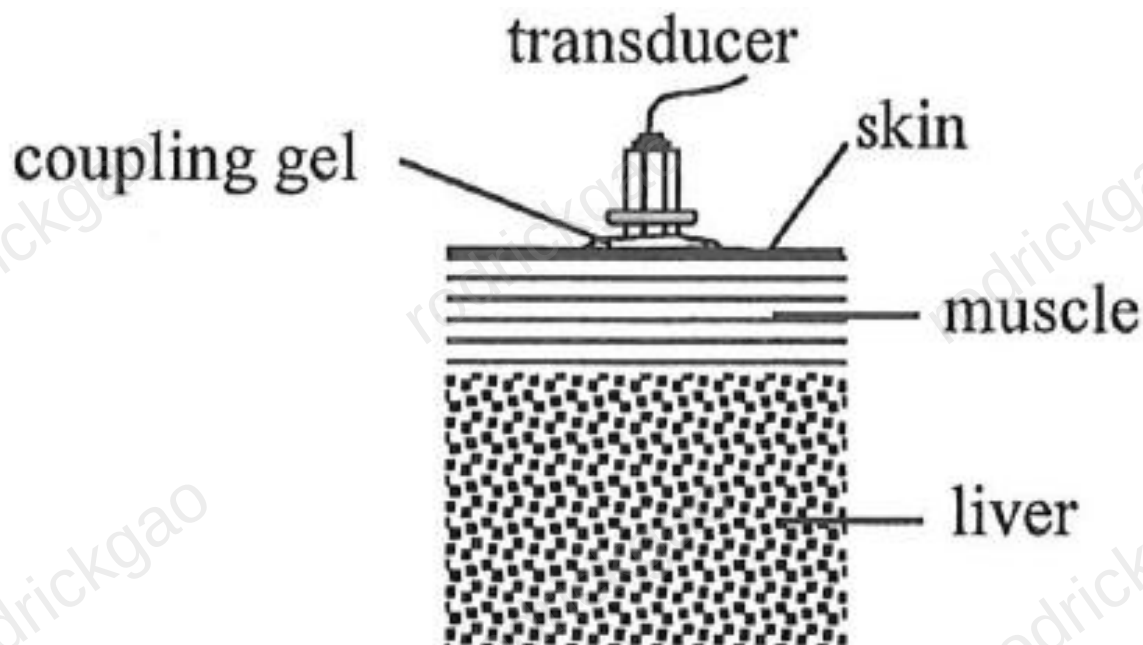
- A. 0.0008°
- B. 0.008°
- C. 0.08°
- D. 0.8°

Official answer: B

$\theta = 1.22\lambda/D = 1.22(550 \times 10^{-9})/(5 \times 10^{-3}) = 1.34 \times 10^{-4} \text{ rad} = 0.0077^\circ \approx 0.008^\circ$. Answer: B.

Question 4.3

An ultrasound transducer is set up as shown below.
ultrasound



Question 4.3 ultrasound transducer

The acoustic impedance of various tissues and the coupling gel used are given below.

acoustic impedance / $\times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$

coupling gel 1.45 muscle 1.71 liver 1.65

Estimate the proportion of the intensity of the incident ultrasound beam that enters the liver from the coupling gel.

A. 0.9929

B. 0.9932

C. 0.9997

D. 0.0071

Official answer: A

Across gel–muscle then muscle–liver, $T = [1 - ((1.71 - 1.45)/(1.71 + 1.45))^2][1 - ((1.65 - 1.71)/(1.65 + 1.71))^2] = 0.9929$. Answer: A.

Question 4.4

Which of the following statements about ultrasound scan is/are correct?

- (1) Performing an ultrasound scan does not require great skills and experience.
- (2) Ultrasound cannot penetrate bone, so it is difficult to image the human brain.
- (3) Lung imaging is difficult as there are air spaces in the lung.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: D

Ultrasound scanning requires operator skill; bone is difficult to penetrate and air in the lungs causes strong reflection. Statements (2) and (3) are correct. Answer: D.

Question 4.5

Which of the following statements about X-ray radiographic imaging is/are correct?

- (1) A radiation dose is cumulative.
- (2) The resolution of soft tissues is good using high energy X-rays.
- (3) The use of contrast media can be unpleasant.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: C

Radiation dose is cumulative; high-energy X-rays reduce soft-tissue contrast; contrast media may cause discomfort. Statements (1) and (3) are correct. Answer: C.

Question 4.6

The half-value thickness of a cobalt-60 γ source (1 MeV) is 12.5 mm in lead and 60.5 mm in concrete. Estimate the minimum thickness of a lead sheet to replace a concrete wall of 25.0 cm thick for radiation shielding of this cobalt-60 source.

- A. 6.1 cm
- B. 5.2 cm
- C. 4.0 cm
- D. 3.1 cm

Official answer: B

Equal attenuation requires the same number of half-value layers: $x_{\text{Pb}}/12.5 = 250/60.5$, giving $x_{\text{Pb}} = 51.7 \text{ mm} = 5.2 \text{ cm}$. Answer: B.

Question 4.7

During a computed tomography (CT) scan, an X-ray tube rotates around the patient and emits X-rays while detectors collect the intensity of X-ray transmission through the patient. Which of the following is the correct sequence of further processing?

- (1) Attenuation data are put through a reconstruction algorithm.
- (2) The intensity of X-ray transmission is converted to digital data.
- (3) Digital data are processed to obtain attenuation data.
- (4) The reconstructed image is available for viewing on a digital display.

A. (3), (1), (4), (2)

B. (2), (4), (3), (1)

C. (3), (1), (2), (4)

D. (2), (3), (1), (4)

Official answer: D

Correct sequence: digitise transmitted intensity → obtain attenuation data → apply reconstruction algorithm → display the image, i.e. (2)(3)(1)(4). Answer: D.

Question 4.8

A tracer has a biological half-life of 9.45 days and an effective half-life of 2.30 days. Find its physical half-life.

A. 1.85 days

B. 7.15 days

C. 3.04 days

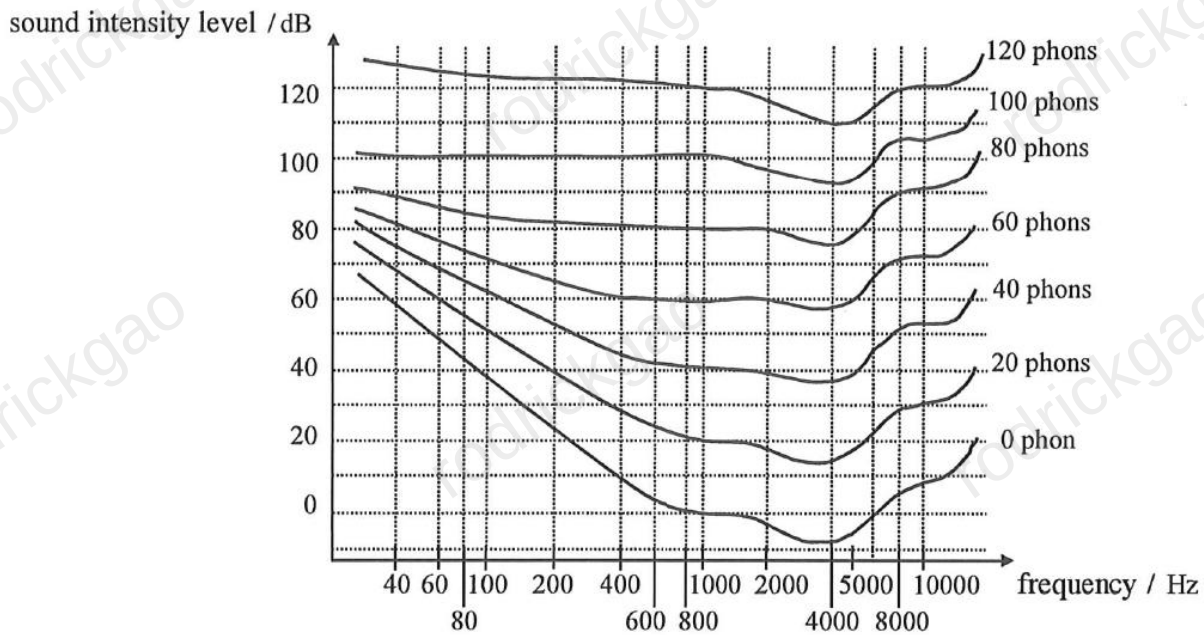
D. 11.8 days

Official answer: C

$1/T_e = 1/T_p + 1/T_b$. Thus $1/T_p = 1/2.30 - 1/9.45$, giving $T_p = 3.04$ d. Answer: C.

Structured Question 4

The figure shows the curves of equal loudness in the phon scale for persons with normal hearing.



Equal-loudness curves

- (a) (i) Explain why they are called ‘curves of equal loudness’. (1 mark)
- (ii) What is the physical significance of the 0 phon curve? (1 mark)
- (b) (i) Which of the following in the ear — the three ossicles, the hair sensory cells and the auditory nerve — is responsible for the generation of nerve impulse for hearing? (1 mark)
- (ii) From what feature of these curves does one know that the ear is most sensitive to sound between 2000 Hz and 5000 Hz? (1 mark)
- (c) (i) At what frequency are loudness and the sound intensity level numerically the same? (1 mark)
- (ii) When the loudness changes from 20 phons to 40 phons, determine whether the change of sound intensity level (in dB) at 60 Hz is the same as that at 8 kHz. Show your working. (2 marks)
- (d) (i) Sounds above 120 phons are painful as well as damaging. Calculate the ratio of intensities of sound between 120 phons and 0 phon at 1000 Hz. (2 marks)
- (ii) How would these curves change for people who suffer from hearing loss mostly for sounds of high frequencies? (1 mark)

Official marking scheme

Solution		Marks	Remarks
4.	(a) (i) Any sound along a given / the same curve is perceived as equally loud by the average person.	1A	
		1	
	(ii) The 0 phon curve corresponds to the threshold of hearing.	1A	
		1	
	(b) (i) The hair sensory cells	1A	
		1	
	(ii) The loudness curves all have minimums (sound intensity level) in them between 2000 Hz and 5000 Hz, which means that the ear is most sensitive to frequencies within that range.	1A	
		1	
	(c) (i) At 1000 Hz (1 kHz)	1A	
		1	
	(ii) Not the same: At 60 Hz, $\Delta L \approx (70 - 60) = 10$ dB At 8 kHz, $\Delta L \approx (50 - 30) = 20$ dB	1A	
		1A	
	(d) (i)	2	
		$120 = 10 \log \left(\frac{I_{120}}{I_0} \right)$	
$\frac{I_{120}}{I_0} = 10^{12}$ (or $\frac{I_0}{I_{120}} = 10^{-12}$)		1A	
2			
(ii) The right end (high frequencies) of the curves become higher / raised.	1A	Accept drawings	
	1		

- (a) (i) Sounds on the same contour are perceived by an average listener as equally loud.
(ii) The 0-phon contour is the threshold of hearing.
(b) (i) Hair cells.
(ii) The required sound intensity level is lowest at 2–5 kHz, so the ear is most sensitive there.
(c) (i) 1000 Hz.
(ii) From the graph: at 60 Hz, $70 - 60 = 10$ dB; at 8 kHz, $50 - 30 = 20$ dB, so the changes differ.
(d) (i) At 1000 Hz, phon = dB; $I_{120}/I_0 = 10^{120/10} = 10^{12}$.
(ii) For high-frequency hearing loss, draw the right-hand end of the contour higher.